



# Shopper Patterns and Analytics Research Kit (SPARK)

Group P22:

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# Problem Statement & Objectives

## Challenge

- E-commerce platforms struggle to identify shoppers likely to purchase, leading to inefficient marketing.

## Objectives

- Predict purchase intent using Logistic Regression and Decision Tree
- Discover buying patterns with association rules
- Present insights through a Streamlit dashboard

## Target Users

- Analysts, business stakeholders, researchers, and students



# Dataset & Class Imbalance



**Source:** UCI Online Shoppers Purchasing Intention Dataset

**Initial Total:** 12,330 sessions

**After cleaning:** 12,205 sessions (125 duplicates removed, 1.01%)

**Total Attributes:** 18 (14 numerical, 4 categorical)

## Class Distribution (After Cleaning)

- **Non-Revenue (False):** 10,297 sessions (84.37%)
- **Revenue (True):** 1,908 sessions (15.63%)
- **Class Ratio:** 1 : 5.4 (Revenue : Non-Revenue)



# Methodology Framework

## Stage 1: Preprocessing

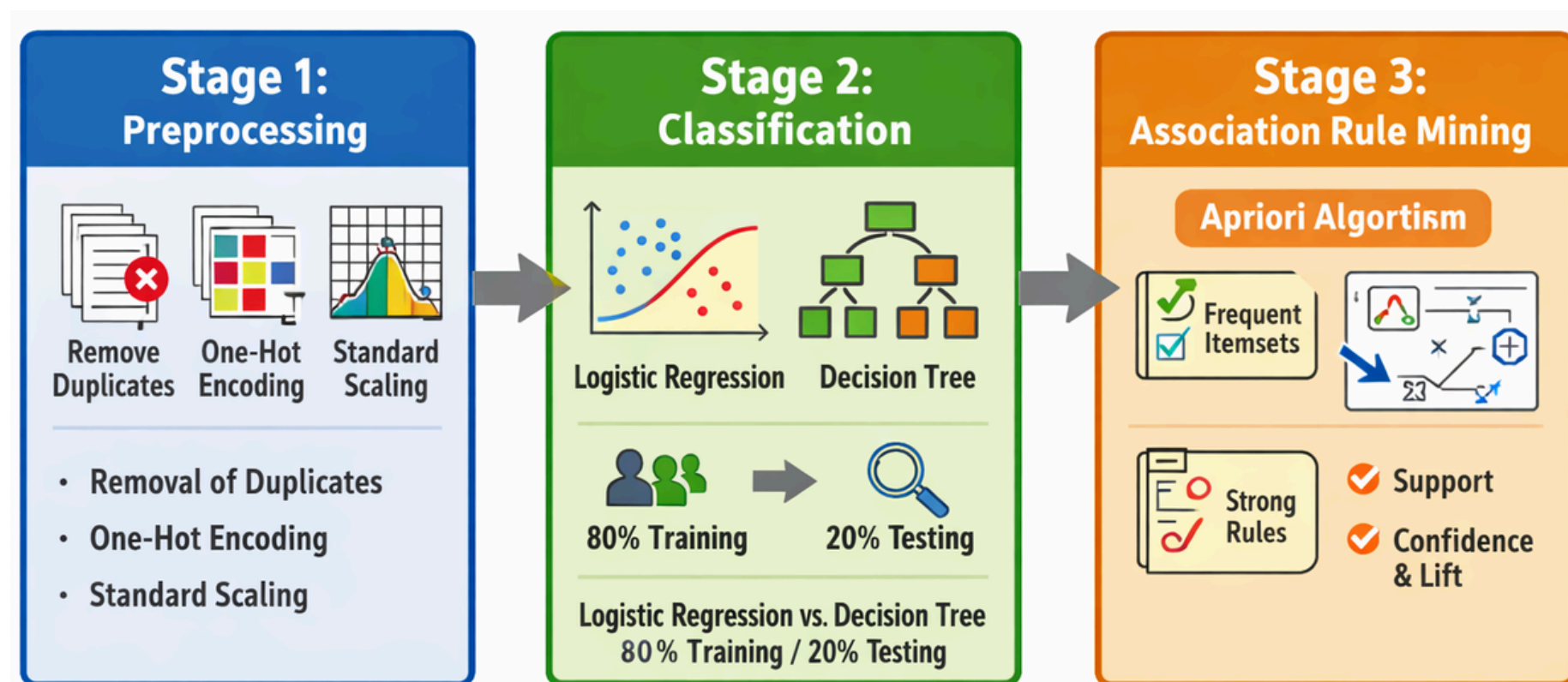
- Removal of duplicates; One-Hot Encoding for categorical variables.
- Scaling: StandardScaler applied to numerical features

## Stage 2: Classification

- Models: Logistic Regression (Linear) vs. Decision Tree (Non-linear).
- Split: 80% Training / 20% Testing.

## Stage 3: Association Rule Mining

- Algorithm: Apriori.
- Goal: Extract frequent itemsets (Support) and strong rules (Confidence/Lift)



# Classification Results

## Models Evaluated

- Logistic Regression
- Decision Tree

## Better Model

- Decision Tree

## Evaluation Metrics

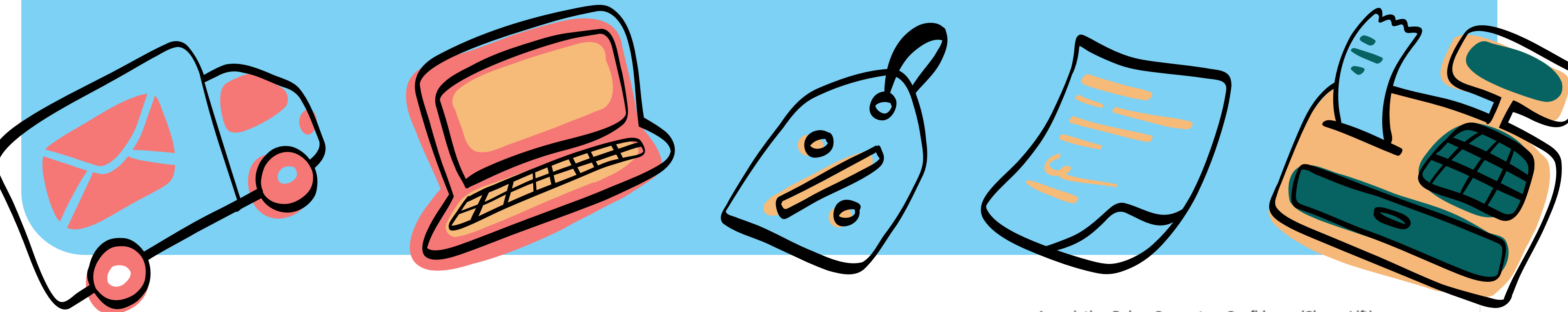
- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

## Parameter Used

- Logistic Regression
  - Max iteration: 1000
- Decision Tree
  - Max depth: 10
  - Min sample split: 50

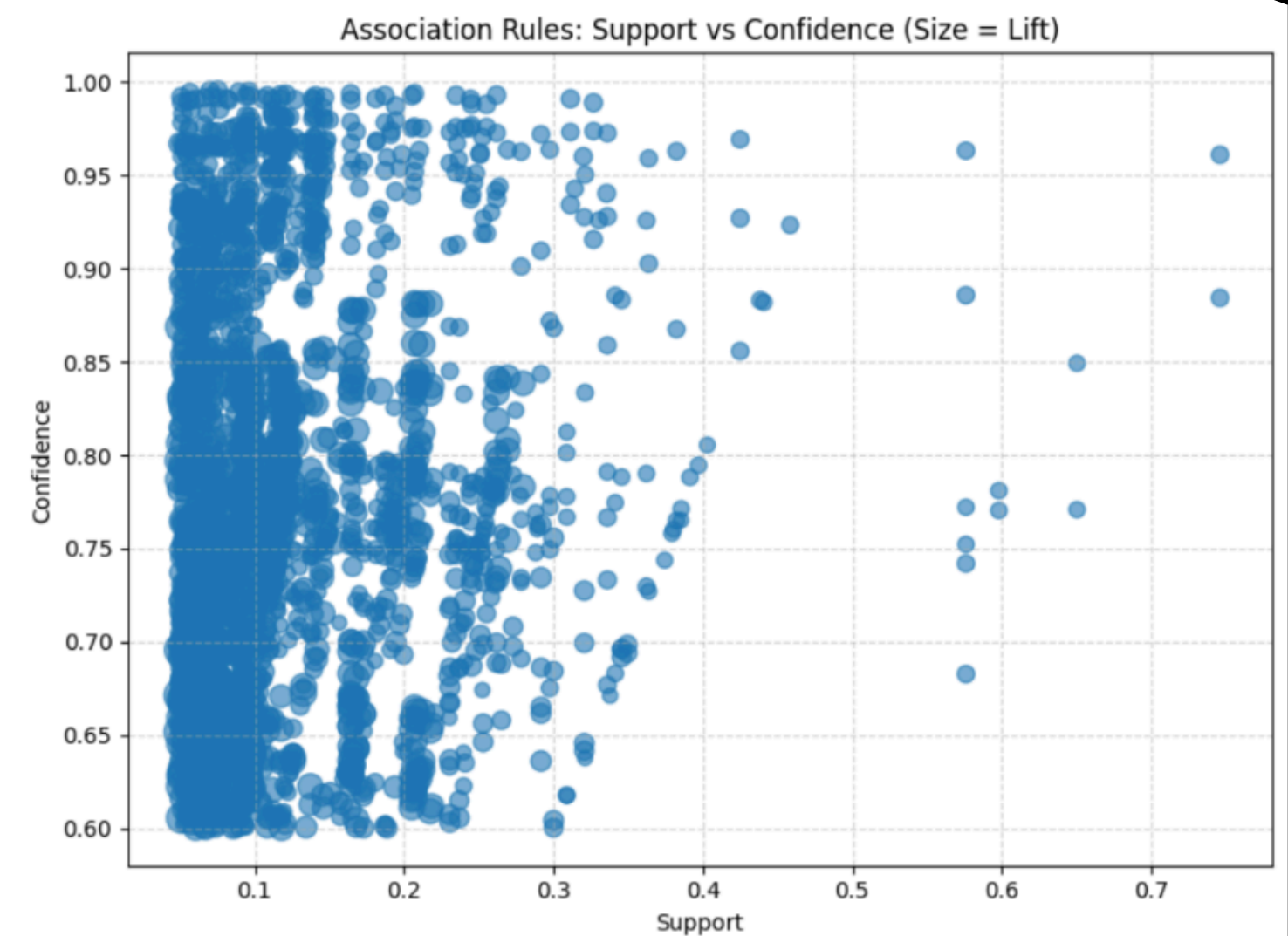
|   | Metric    | Logistic Regression | Decision Tree | Difference (LR - DT) | Better Model        | % Difference (LR vs DT) |
|---|-----------|---------------------|---------------|----------------------|---------------------|-------------------------|
| 0 | Accuracy  | 0.8894              | 0.8902        | -0.0008              | Decision Tree       | -0.09                   |
| 1 | Precision | 0.7718              | 0.6500        | 0.1218               | Logistic Regression | 18.74                   |
| 2 | Recall    | 0.4162              | 0.6466        | -0.2304              | Decision Tree       | -35.63                  |
| 3 | F1-Score  | 0.5408              | 0.6483        | -0.1075              | Decision Tree       | -16.58                  |
| 4 | ROC-AUC   | 0.9022              | 0.9080        | -0.0058              | Decision Tree       | -0.64                   |





# Association Rule Mining Findings

- Apriori algorithm was used to discover frequent shopper behavior patterns
- High product engagement is strongly associated with purchase outcomes
- Returning visitors with high page value show higher likelihood of conversion
- High bounce rate and short sessions are linked to non-purchase behavior
- Temporal factors (weekend, special days) strengthen purchase-related rules

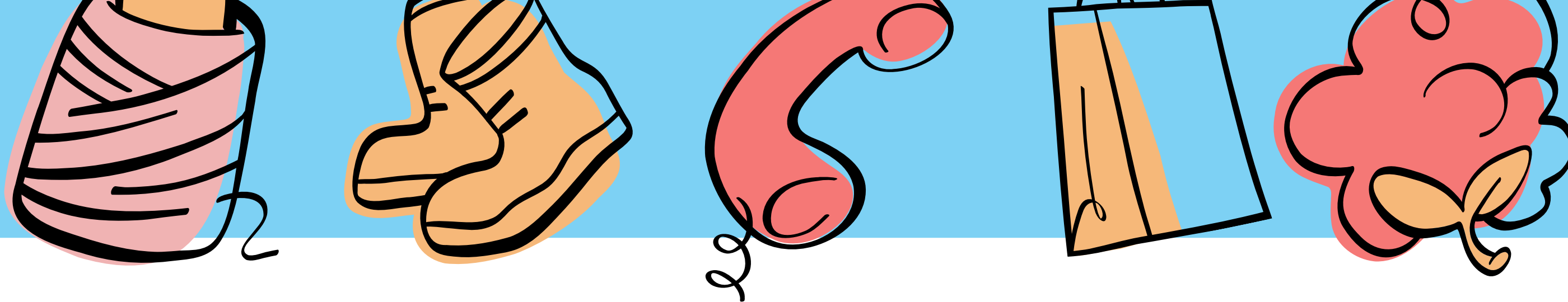


Each point represents a discovered rule; larger points indicate stronger lift.

# The Solution: SPARK Dashboard

- Overview: An interactive web application built with Streamlit.
- Main Function:
  - Data Explorer: Visualize distributions and correlations.
  - Model Prediction: Input real-time session data (Duration, Page Types) to get a Purchase Probability score.
  - Association Rules: Filter and explore behavioral rules by Confidence and Lift.





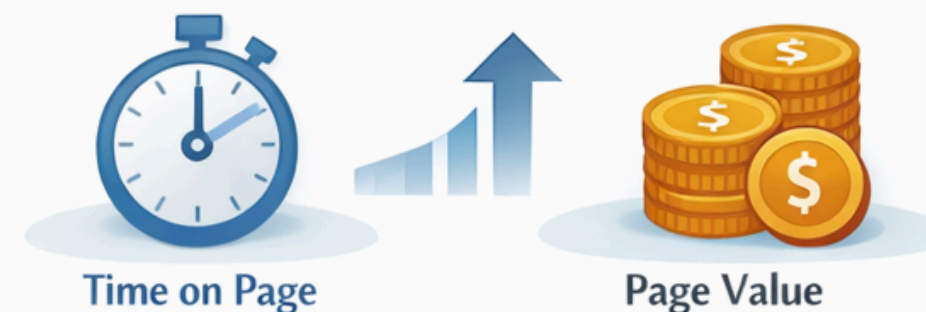
# Key Insights & Conclusion

Decision Tree model performs better  
— at identifying actual buyers —



Decision Tree model performs better at  
identifying actual buyers  
(Higher Recall)

Longer time on product pages and higher page value  
— strongly indicate purchase intent —

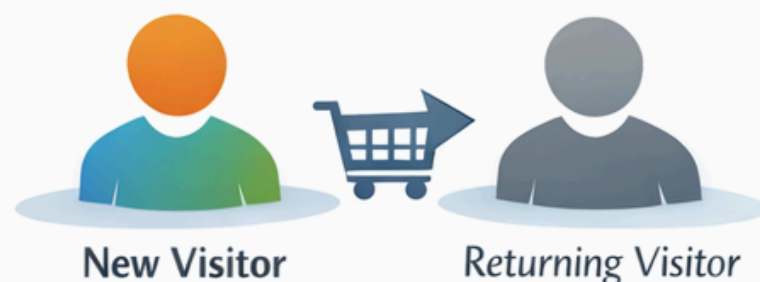


Time on Page

Page Value

Longer time on product pages and higher page value  
strongly indicate purchase intent

New visitors show a higher likelihood  
— of making a purchase —



New Visitor

Returning Visitor



Purchasing activity peaks in November  
— due to major sales events —



# Thank you

- Explore the Dashboard
- Q&A

